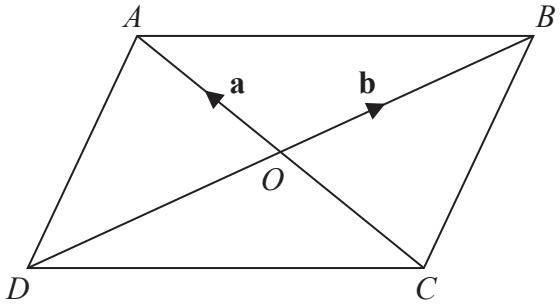
27

Diagonals bisect each other.

DO = OB = b



ABCD is a parallelogram.
The diagonals of the parallelogram intersect at *O*.

$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$

(a) Find, in terms of **b**, the vector $\overrightarrow{DB}$.

DB = DO + OB = b + b = 2b

1 mark
Two copies of b

2b

(1)

(b) Find, in terms of **a** and **b**, the vector $\overrightarrow{AB}$.

AB = AO + OB = −a + b

1 mark
Reverse OA, then add OB

b − a

(1)

(c) Find, in terms of **a** and **b**, the vector $\overrightarrow{AD}$.

AD = AO + OD = −a − b

1 mark
Both directions reversed

−a − b

(1)

(Total for Question 27 is 3 marks)

TOTAL FOR PAPER IS 80 MARKS

DO NOT WRITE IN THIS AREA

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